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SERIES OF TECHNICAL NOTES ON THE IMPACT OF NATURAL GAS DEFICIT AND THE INCREASE IN PRICES FOR END USERS: GENERAL PRESENTATION

Note 1. Increase in the price of residential natural gas in scenarios of higher imports

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Technical notes series on the impact of the natural gas shortage and price increases for end users: general overview

Due to a combination of past and present public policies, which have introduced transaction costs to the exploration and exploitation of conventional and unconventional domestic natural gas, indecisions on the use, size, location and remuneration of new import facilities, inflexibilities in contracting and charging by distance of the transport system, Colombia faces a deficit and possible shortage of natural gas in the short term.

Fedesarrollo presents four technical notes on the impacts of the natural gas deficit due to the reduction in domestic supply, the difficulties in expanding import capacities in the short term, and the increase in prices for end users due to import needs.
These notes aim to provide elements for decision-making within the State and to inform the public with public data and arguments guided by the general interest.

The titles of the technical notes are as follows:

Note 1. Increase in the price of residential natural gas in scenarios of greater imports.

Note 2. Fiscal cost of increased subsidies due to rising natural gas prices.

Note 3. Macroeconomic costs of reducing the supply of natural gas.

Note 4. Short and medium term public policy and regulation.

Note 5. Energy poverty due to reduced supply and increased price of natural gas.

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Note 1. Increase in the price of residential natural gas in scenarios of higher imports

Sergio Cabrales and Juan Benavides (Fedesarrollo, Colombia)1 .

Objective and main message

Over the past eleven years, Colombia's proven natural gas reserves have been reduced by more than half, falling from 5,727 gigacubic feet (GCF) in 2012 to 2,373 GCF in 2023. During this period, with the exception of 2021, new reserves have been insufficient to replace the volume of gas produced. Due to the narrowing gap between supply and demand, the price of gas has risen from US$4.5 per million BTU (MBTU) between 2015 and 2022 to over US$10 per MBTU in recent months, not including the final price which incorporates the remaining costs in the supply chain.

Projections of the supply and demand balance, according to official information, indicate that by the end of 2026, the supply of domestic and imported natural gas will not be sufficient to meet the demand projected by the UPME (Mining and Energy Planning Unit). Imported gas, in addition to being insufficient to cover the local deficit due to the size of the regasification plant on the Atlantic Coast, has prices that can be up to three times higher than those of domestic gas.

In this context, this document estimates the increase in the natural gas tariff for the residential sector, considering different levels of imported gas participation. The percentage increase in the natural gas tariff (variable unit cost, VUC) will be calculated in five of the main cities, compared to the average tariff observed in 2024.

If half of the supply came from imported gas, the tariff increase could reach 44.6% in Bogotá, 44.9% in Medellín, 45.8% in Bucaramanga, 30.3% in Cali, and 11.7% in Barranquilla, respectively. In a scenario where the supply consists exclusively of imported gas, the increases could be 89.3% in Bogotá, 89.7% in Medellín, 91.5% in Bucaramanga, 60.7% in Cali, and 23.4% in Barranquilla.

Colombia is ceasing to be an attractive destination for hydrocarbon investment. Current conditions in the hydrocarbon sector have led to the withdrawal of major oil companies such as ExxonMobil, Shell, Repsol, ConocoPhillips, Chevron, and BP, which no longer have any significant projects planned for the country. The domestic production deficit was avoidable if, a decade ago, the environmental licensing and prior consultation processes had been handled swiftly with public leadership, if additional investment decisions had been made regarding imports, and if, after 2022, obstacles to exploration had not been introduced and *government intervention had not been increased.*

1 Disclaimer: The opinions expressed in this publication are those of the authors and should not be attributed to Fedesarrollo.

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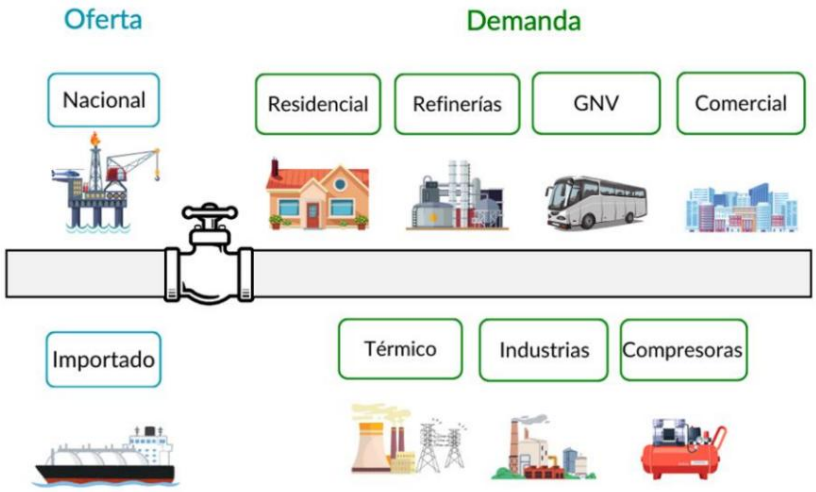
In a critical scenario, if no new gas supply is added by the end of 2026, the total availability of natural gas—both domestic and imported—could fall short of domestic demand. This situation could lead to gas rationing in some sectors or even nationwide power outages. Technical notes 3 and 4 of this series will illustrate the macroeconomic impact of reduced supply and its effect on energy poverty.

Industrial structure and regulation

With the discovery of the Chuchupa and Ballena gas fields during the 1970s, and the Cusiana field in the 1990s, the use of natural gas became widespread in the industrial, residential, and even vehicular (CNG) sectors. Of particular importance was the replacement of hazardous fuels such as kerosene and firewood with natural gas, especially in lower-income residential areas, and the substitution of liquid fuels and coal in industry. Natural gas is an energy source with significant social benefits and supports productivity in Colombia.

During 1994 and 1995, the natural gas market was structured and its regulatory framework was established. Law 142 of 1994 on residential public services included the combustible gas supply chain, with the links presented in Figure 1. In 2023, natural gas represented 14.5% (222 PJ) of final energy consumption (1529 PJ), in a context where diesel (322 PJ), gasoline (323 PJ), and electricity (250 PJ) accounted for 58.5% of total consumption. 2

Figure 1. Supply chain: production or import, transport, and distribution/ marketing



Source: Fedesarrollo's own elaboration.

The natural gas industry in Colombia is structured into three main segments: (i) exploration and production; (ii) transportation; and (iii) distribution and marketing. This chain is regulated by entities such as the UPME, the CREG, and the Ministry of Mines and Energy, and has

2 UPME (2023). Colombian Energy Balance (BECO). https://www1.upme.gov.co/DemandayEficiencia/Paginas/BECO.aspx

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Public and private participation. Ecopetrol and Canacol Energy, as the main producers, along with transport companies such as TGI and Promigas, and regional distributors, coordinate the supply at the national level.

In the case of domestically produced natural gas, price formation follows a regulated market system with freedom of negotiation between the parties, in effect since 2013. Under this system, prices are determined primarily by the delivery point, contractual conditions (such as terms, volumes, and supply firmness), and the dynamics of supply and demand in the domestic market. To ensure transparency and efficiency, the Colombian market employs mechanisms such as supply auctions.

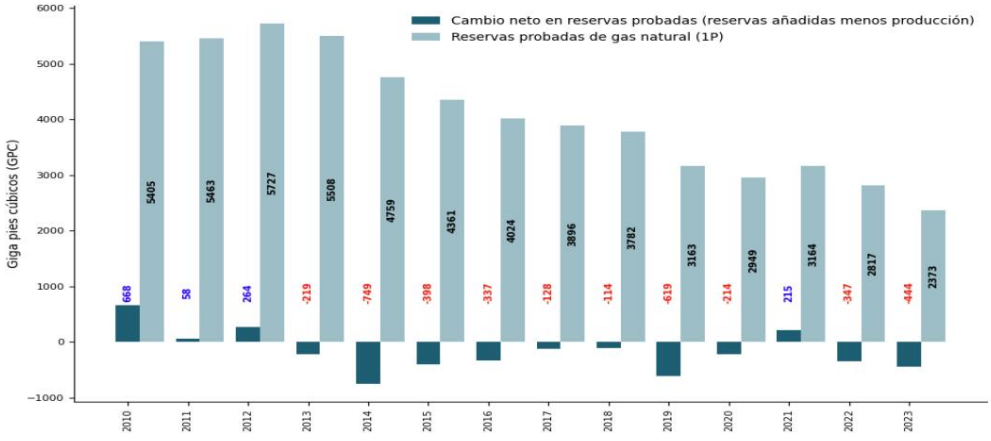
The price of imported gas—in the form of liquefied natural gas (LNG)—is typically indexed to international benchmarks such as the *Henry Hub* (United States). Added to this price are the costs of liquefaction, maritime transport, regasification at import terminals, and port charges. As a result, the price of imported gas tends to be significantly higher than that of domestically produced gas.

Natural gas tariff regulation is based on Law 142 of 1994, which establishes the general framework for residential public services, and is further developed through numerous resolutions issued by the Energy and Gas Regulatory Commission (CREG). This legislation covers key aspects such as the definition of subsistence consumption (CREG Resolution 057 of 1996), the application of differentiated contributions based on user type (CREG Resolutions 124 and 015 of 1996 and 1997), and the criteria for remunerating distribution and marketing activities (CREG Resolutions 202 of 2013 and 102003 of 2022). In addition, general tariff formulas have been established for regulated users (CREG Resolution 137 of 2013).

Reduction in the national supply of natural gas

Despite its significant geological potential, Colombia has experienced a marked reduction in the addition of proven gas and oil reserves. In the last thirteen years, Colombia's proven natural gas reserves have been reduced to less than half (Figure 2).

Chart 2. Net change in proven reserves and total proven reserves (2010-2023)



Source: Prepared by Fedesarrollo with data from the National Hydrocarbons Agency (ANH).

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During this period, with the exception of 2021, the reserves added have been insufficient to replace the volume of gas produced, which explains the constant decrease in proven reserves.

The decline in domestic supply is due to a combination of six causes:

- **C1. Exploration risk.** Between 1952 and 2010, the exploration success factor has been 23%, which means that only one in four wells drilled achieves commercial viability.3 This indicator reflects the risks associated with hydrocarbon exploration in the country.

- **C2. Increased state participation in oil revenues.** Until 2018, incentives were provided
  Investment through deductions and tax benefits consolidated the sector as an economic engine of the country. The 2022 tax reform increased the State's share of the profits generated by hydrocarbon projects, raising the *government take* from 63% with the imposition of surcharges (with additional rates of 5%, 10% or 15%), to more than 80%.

  (depending on international oil prices), higher dividend taxes, and restrictions on asset amortization. This tax-driven approach drastically reduces the profitability of investments and discourages investment in new exploration and projects.
  of hydrocarbons.

- **C3. Reduction and slowdown in exploration activity.** Historically, Colombia has offered more than 1,000 blocks, with 413 contracts signed.6 While between 2010 and 2014, between 110 and 130 wells were drilled annually, that number dropped to 34 in 2024.7 This low activity limits new discoveries and the addition of proven reserves. Furthermore, since 2021, the country has not awarded any new blocks for exploration and production, and the current government has been clear in its decision not to sign any additional contracts.

- **C4. Complexity of environmental licensing, social licensing, and prior consultation procedures.** These procedures take years to complete. The regulatory environment in these matters prolongs execution times, increases project costs, and reduces profitability by delaying the start of commercial operation.

- **C5. Increased insecurity.** In the last two years, Colombia has registered more than 2,708
  Blockade events and nearly 69 attacks on oil infrastructure. [8] This situation implies
  risk to operators, disruptions in field operations and increases in the

3 López-Enciso, EA, Montes-Uribe, E., Garavito-Acosta, AL, & Collazos-Gaitán, MM (2012). The oil economy in Colombia (Part I): legal-contractual framework and main links in the production chain (1920-2010).

4 The *government take* is the part of the real economic rent generated by an investment during the life of a project that a government charges, calculated in present value of the cash flows it will receive.

5 Corficolombiana. (2022, October 18). *Special report: Tax reform presentation 2022.* https://investigaciones.corficolombiana.com/documents/38211/0/221018-Informe-Especial-Ponencia-Reforma-tributaria- 2022vf%20(1).pdf/fca94e31-9742-026a-7694-c7e7dd4aedf2

6 National Hydrocarbons Agency. (2021). Results of the Colombia Round 2021. https://anh.gov.co/es/mod-de-rondas/resultados-ronda-colombia-2021/

7 Campetrol (2025). Drilling and Production Report. https://campetrol.org/informe-de-taladro-y-produccion/

8 Portafolio (2025). Blockades against oil operations remain on the rise. https://www.portafolio.co/energia/los-bloqueos-contra-la-operacion-petrolera-se-mantienen-disparados-627315 Campetrol (2025). Drilling and Production Report. and https://campetrol.org/informe-de-taladro-y-produccion/

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operating costs due to increased expenses in security, infrastructure repairs, and insurance premiums.

- **C6. Concentrated industry structure and regulatory weakness.** The production market is highly concentrated (Ecopetrol controls more than 80% of domestic supply)
  This reduces competitive pressure on supply. Furthermore, the expansion and charging for transportation based on distance in a poorly connected network discourages the entry of highly productive fields, and prejudice has hampered the entry of *fracked gas.*

Colombia is ceasing to be an attractive destination for hydrocarbon investment.
Currently, of the 302 active hydrocarbon exploration and production contracts in Colombia, 39 are suspended and 57 are in the process of termination, meaning that more than 30% of the contract portfolio is inactive.⁹
These suspensions are mainly due to social conflicts, environmental licensing issues, or public order concerns, such as the presence of illegal armed groups. The current conditions in the hydrocarbon sector have led to the withdrawal of major oil companies such as ExxonMobil, Shell, Repsol, ConocoPhillips, Chevron, and BP, which no longer have significant projects planned in the country.¹⁰

Discoveries in the deep waters of the Caribbean Sea, such as those made by Ecopetrol in partnership with Petrobras in the Sirius project, have revealed high gas potential that could double current proven gas reserves (2.3 Tcf). Sirius could produce more than 450 million cubic feet per day (MMcf/d), equivalent to approximately 40% of national demand.

Currently, however, its development faces significant technical, logistical, and regulatory challenges. Added to this is a complex social landscape where more than 100 prior consultation processes must be completed before obtaining the environmental license for the 117-kilometer underwater gas pipeline, which would transport gas from the deep-water field to the Ballena field in La Guajira.

Increase in imported gas since 2023

Before the construction of the Antonio Ricaurte gas pipeline between Venezuela and Colombia in 2008, Colombia's natural gas market was completely closed. Although the project envisioned the export of Colombian gas to Venezuela for the first four years and, subsequently, imports from that country for a period of sixteen years, this latter commitment by the Venezuelan government was never fulfilled. To strengthen security of supply and the reliability *of the electrical system,* Colombia integrated itself into the international Liquefied Natural Gas (LNG) market in 2016 through the commissioning of the terminal at

9. A third of oil contracts are suspended or being terminated; the most affected regions. El Tiempo. https://www.eltiempo.com/economia/sectores/estos-son-los-departamentos-donde-mas-se-dificulta-la-operacion-de-las-petroleras-3326137

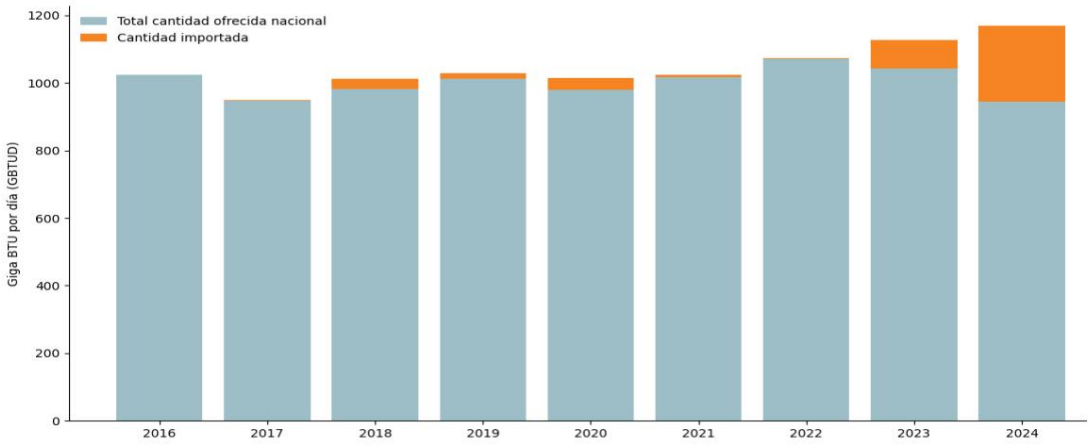
10 Bloomberg Line (2025). *Race for new oil reserves in Latin America leads to record investment.* https://www.bloomberglinea.com/2024/06/21/carrera-por-nuevas-reservas-de-petroleo-en-america-latina-lleva-a-una-inversion-record/

11 Ecopetrol (2025). *He concluded with success the phase of evidence of Sirius-2.* https://www.ecopetrol.com.co/wps/portal/Home/es/noticias/detalle/concluyo-con-exito-la-fase-de-pruebas-de-sirius-2

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Imports from Sociedad Portuaria El Cayao SA ESP (SPEC LNG), located in Cartagena. The total supply of gas, both domestic and imported, has the evolution shown in Graph 3.

Figure 3. Average annual supply of domestic and imported gas (2016-2024)



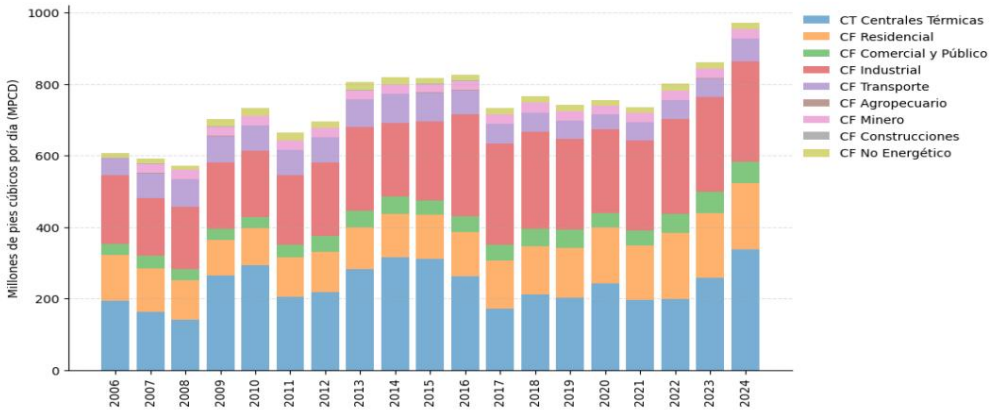
Source: Prepared by Fedesarrollo with data from UPME, Bolsa Mercantil de Colombia, and the Natural Gas Market Manager.

Since the end of 2024, SPEC LNG has not only imported gas for the electricity sector on an occasional basis, but its imports now cover a portion of *essential demand,* including the residential, commercial, and compressed natural gas (CNG) vehicle sectors, as well as refineries and compressor stations (Figure 3). In 2024, 23.8% of the total gas offered was imported, compared to 8.1% in 2023.

Gas demand

In the last two decades, the sectors that demanded the most gas were thermal power plants. (31.4%) and the industrial sector (30.7%), followed by the residential sector (17.9%), the transport sector (8.5%), and the commercial sector (5.7%) (Graph 4).

Figure 4. Natural gas demand by sector (2006–2024)



Source: Prepared by Fedesarrollo with data from the Colombian Energy Balance (2006-2023) of UPME, Natural Gas Market Manager and CAMPETROL.

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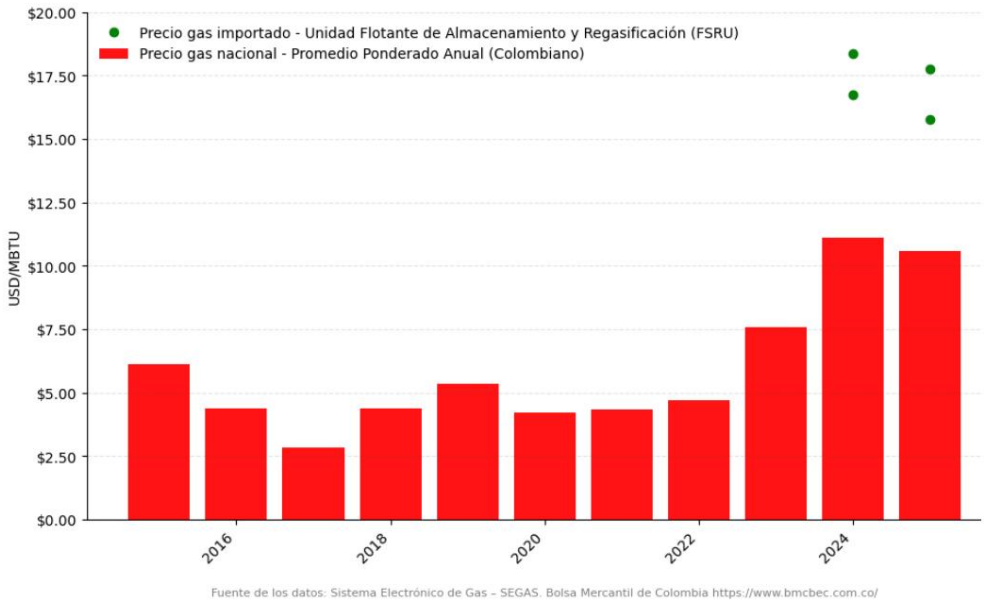
Gas consumption by thermal power plants increases during periods of low hydrology or during the occurrence of the El Niño phenomenon, when hydroelectric generation decreases.
In terms of growth rates, the commercial sector saw the largest increase in natural gas demand, at 85% during that period, followed by the thermal sector, which experienced a 73% increase. Meanwhile, the residential and industrial sectors each saw growth of approximately 47%.

Currently, all Colombians use natural gas directly or indirectly.
Approximately 10% of the electricity consumed in the country is generated by thermal power plants. More than 11 million households use natural gas for cooking. In addition, sectors such as refining, transportation, cement, steel, food processing, paper, and mining also use it in their production processes. Due to its versatility, cost, and lower environmental impact compared to other fossil fuels, gas is an essential component of the country's energy mix.
It is an engine for productivity in various sectors and is an energy source with a high social impact.

Production price (G)

The price of natural gas in Colombia depends on the origin of the supply (whether domestic production or imports), contractual conditions, and market dynamics.
national and international. Before 2023, average domestic gas contract prices ranged from USD $2.80 to USD $5.40 per million BTU (MBTU) (Chart 5). In 2023, these prices increased to an average of USD $7.50 per MBTU, and have recently exceeded USD $10.00 per MBTU.

Chart 5. Average price of domestic and imported gas contracts - Primary Market (2016-2025)



Source: Prepared by Fedesarrollo with data from the Electronic Gas System (SEGAS), Colombian Mercantile Exchange.

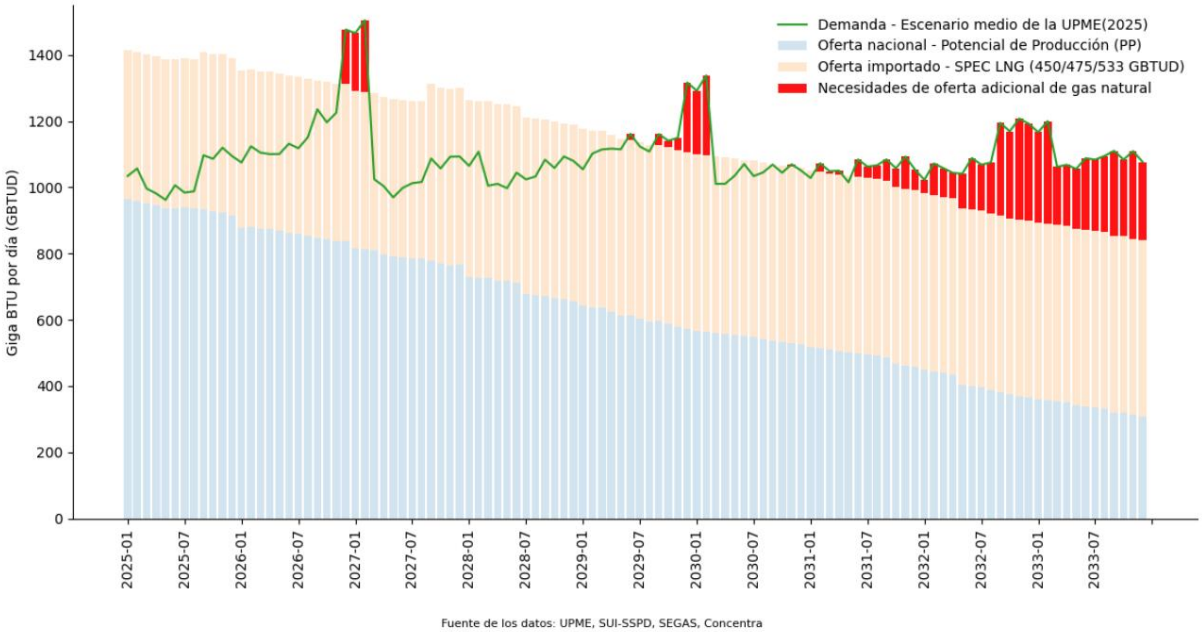
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Recently, the average price of imported gas (green dots in Graph 5) has fluctuated
Between US$15.76 and US$18.39 per MBTU, levels significantly higher than the domestic gas price (red bars). As previously noted, this difference is mainly explained by the additional costs of liquefaction, shipping, and regasification, as well as by purchasing gas on the secondary market due to the lack of long-term contracts on the international primary market.

Balance between domestic and imported supply and demand

After 45 years, on December 1, 2024, Colombia lost its self-sufficiency in meeting the demand for natural gas in the residential, commercial, and vehicular (CNG) sectors, as well as in refineries and compressor stations, with its domestic production. Comparing the national supply (blue bars)
The comparison of domestic and imported natural gas (orange bars) with expected demand (green line) reveals a growing dependence on imported gas in the medium term (Graph 6). Projections of the supply-demand balance, based on official information, indicate that by the end of 2026, the supply of domestic and imported natural gas will not be sufficient to cover the demand projected by UPME (red bars).

Figure 6. National balance between national supply (Production Potential, PP) and imports versus projected demand



Source: Prepared by Fedesarrollo with data from the Ministry of Mines and Energy, UPME, SUI-SSPD, SEGAS, Concentra.

Gas rates for residential users

The natural gas tariff for residential users, known as the Unit Cost of Service Provision (CU), is made up of two elements: (i) the product between the consumption in meters

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cubic meters (m³) and the Variable Unit Cost (VUC), which reflects the cost per cubic meter consumed and includes the cost components of gas molecules (G), transportation (T), and distribution (D); and (ii) the Fixed Unit Cost (FUC), which corresponds to the fixed component charged on each bill, regardless of the volume consumed. 12

The CUV varies according to the geographical area, the origin of the gas (domestic or imported), the transport from the source of production or import to the distributor *(city gate),* and represents the part of the tariff that depends directly on the user's consumption level.

This rate is subject to subsidy or contribution factors based on the user's socioeconomic status, in accordance with Law 142 of 1994 and the resolutions of the Energy and Gas Regulatory Commission (CREG). Users in socioeconomic strata 1 and 2 may receive subsidies of 60% and 50%, respectively, while users in strata 5 and 6, as well as the commercial and industrial sectors, contribute 20% to finance these subsidies.

Price increase for residential users

The increased share of imported gas in the consumption basket—which is considerably more expensive than domestically produced gas—translates into an increase in the component associated with the cost of gas molecules (G), as well as a change in transportation costs (T) due to the shift from the point of origin to the port of import. This raises the final tariff for residential users and poses an additional challenge to the sustainability of the current subsidy scheme.

As the share of imported gas in residential supply increases, there is a progressive increase in the natural gas tariff (CUV), driven mainly by the increased cost of the component corresponding to the cost of the molecules (G), since imported gas is significantly more expensive than domestically produced gas.

Additionally, the transportation component (T) also increases in cities like Bogotá, Medellín, Cali, and Bucaramanga, because transportation costs rise when the supply point shifts from inland fields such as Cusiana, Cupiagua, or Gibraltar to the import port in Cartagena. As the proportion of imported gas in the supply mix grows, a progressive increase in residential tariffs is observed in the country's major cities.

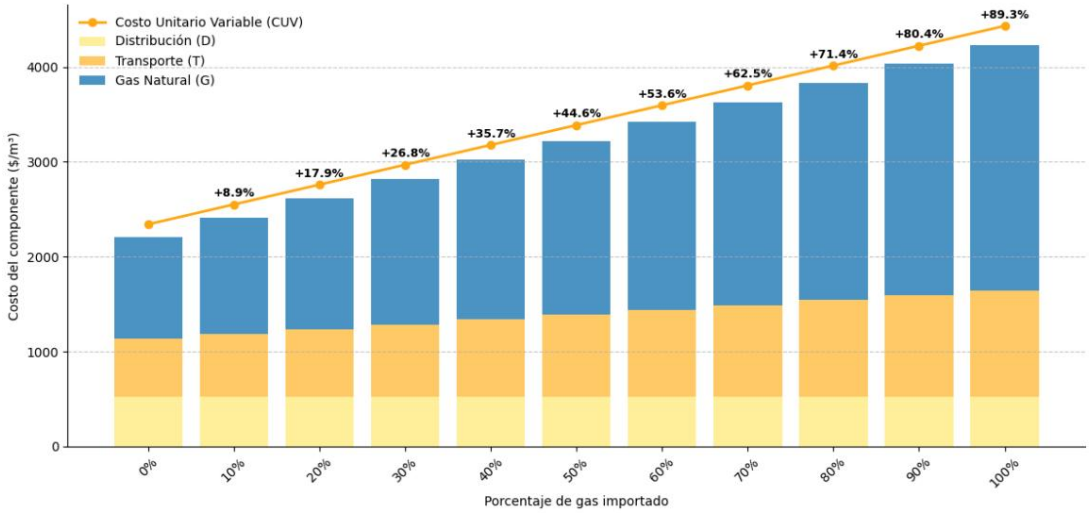
Using supply and price models on the transport network, it is found that, in Bogotá, in scenarios where 50% and 100% of the gas consumed is imported, the rate could increase by up to 44.6% and 89.3%, respectively, affecting more than 2.2 million households (Graph 7).

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12 = (+)/(1 ÿ) + (ÿ) where are the recognized losses and is the calorific value factor.

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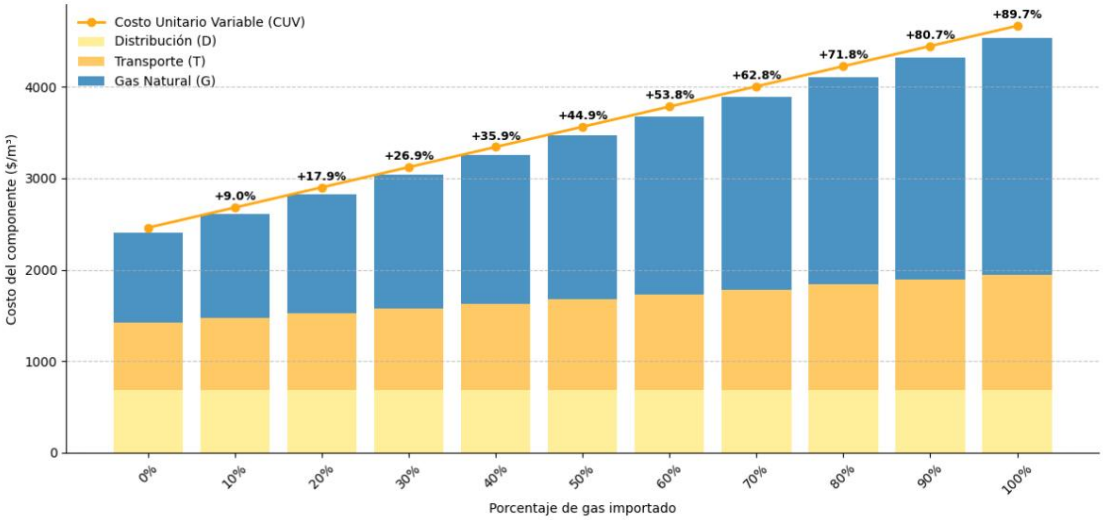
Chart 7. Bogotá – Variation of the natural gas tariff for residential users according to the percentage of imports



Source: Prepared by Fedesarrollo with data from the Superintendency of Public Utilities, Electronic Gas System (SEGAS) and Energy and Gas Regulatory Commission (CREG).

In Medellín, an increase of up to 44.9% and 89.7% is projected in the same scenarios, with an impact on more than 750,000 households that depend on this service (Graph 8).

Chart 8. Medellín – Variation of the natural gas tariff for residential users according to the percentage of imports

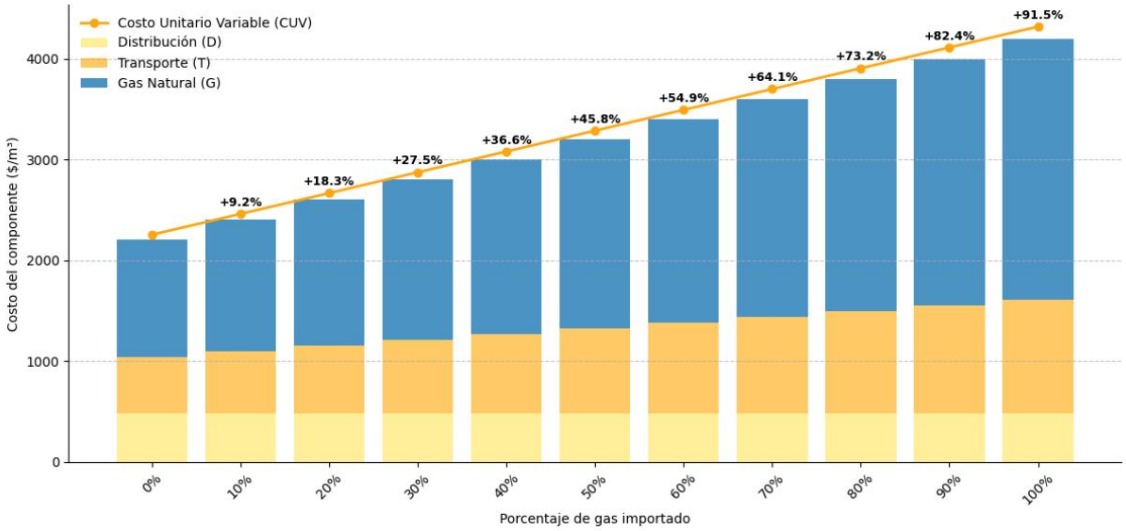


Source: Prepared by Fedesarrollo with data from the Superintendency of Public Utilities, Electronic Gas System (SEGAS) and Energy and Gas Regulatory Commission (CREG).

In Bucaramanga, the increases are even more pronounced, reaching up to 45.8% and 91.5%, respectively, which represents the largest tariff increase among the cities analyzed (Graph 9) and would affect more than 178,000 households.

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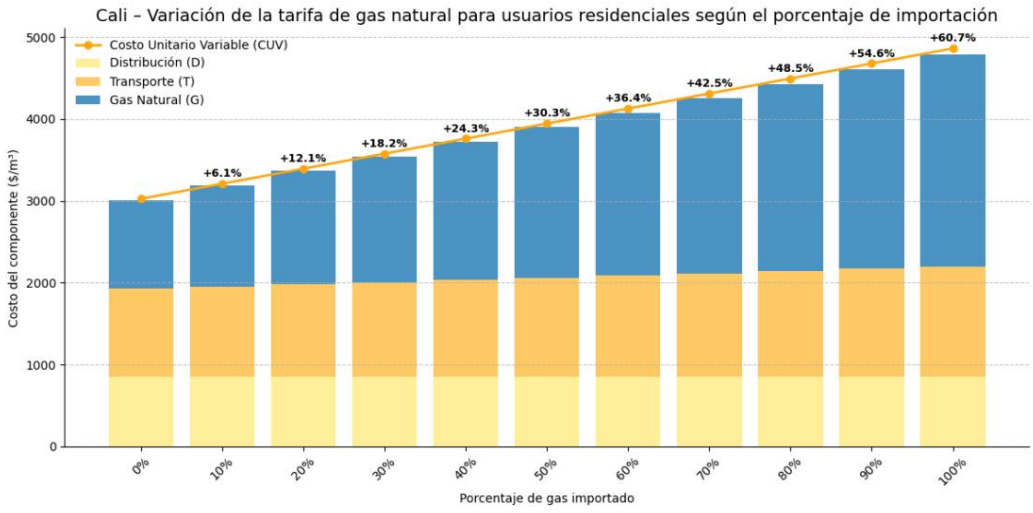
Chart 9. Bucaramanga – Variation of the natural gas tariff for residential users according to the percentage of imports



Source: Prepared by Fedesarrollo with data from the Superintendency of Public Utilities, Electronic Gas System (SEGAS) and Energy and Gas Regulatory Commission (CREG).

In Cali, with a 50% share of imported gas, the rate rises on average by 30.3%, while with 100% imported gas the increase reaches 60.7% (Graph 10), which affects more than 650,000 households.

Chart 10. Cali – Variation of the natural gas tariff for residential users according to the percentage of imports

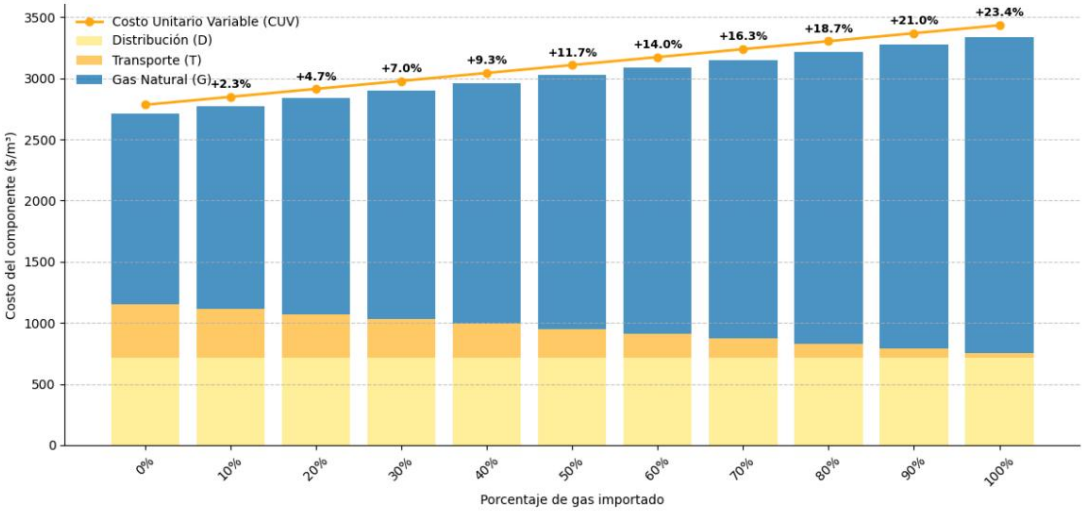


Source: Prepared by Fedesarrollo with data from the Superintendency of Public Utilities, Electronic Gas System (SEGAS) and Energy and Gas Regulatory Commission (CREG).

In contrast, Barranquilla experiences a more moderate tariff impact: with a 50% and 100% share of imported gas, the tariff could increase by 11.7% and 23.4%, respectively (Figure 11), resulting in a relatively low impact on its 368,000 households. This is because, although the cost of the gas itself (G) increases, the effect is partially offset by the reduction in the transportation component (T), given the city's proximity to the Cartagena regasification port.

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Chart 11. Barranquilla – Variation of the natural gas tariff for residential users according to the percentage of imports



Source: Prepared by Fedesarrollo with data from the Superintendency of Public Utilities, Electronic Gas System (SEGAS) and Energy and Gas Regulatory Commission (CREG).

Final considerations

The domestic production deficit was avoidable if, a decade ago, the environmental licensing and prior consultation processes had been handled swiftly with public leadership, if additional investment decisions had been made regarding imports, and if, after 2022, obstacles to exploration had not been introduced and *government take had not been increased.* Consumers pay a surcharge that reduces the competitiveness of the productive sector and the well-being of citizens.

Colombia will not be able to regain its natural gas self-sufficiency without signing new exploration contracts, significantly increasing well drilling, and establishing attractive conditions that stimulate investment in the hydrocarbon sector. Measures to restore the sector's role and strength will be discussed in Note 5 of this series.